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Human-centered AI for teacher educators: Designing professional learning for critical AI literacy

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A B S T R A C T

Teacher educators play a pivotal role in shaping how preservice teachers understand, evaluate, and integrate artificial intelligence (AI) in classrooms, yet they rarely receive systematic preparation to lead this work. This design-based study examines how principles of Human-Centered AI (HCAI) and critical AI literacy can inform the design of professional learning resources for teacher educators. Across three cycles of needs analysis, prototyping, and refinement with seven teacher educators, findings identified three convergent design needs: (a) improving instructional efficiency and effectiveness while preserving professional judgment, (b) modeling responsible and ethical AI integration, and (c) reforming teacher education pedagogies with AI. We translate these findings into a five-module curriculum that operationalizes critical AI literacy competencies (foundational AI knowledge; ethics and algorithmic bias; pedagogical integration; implementation, guidelines, and policy; and human-centered AI in education) through seven HCAI-informed activities, including educator-in-the-loop tasks that strengthen professional judgment, transparency, and equity-oriented decision-making. The study identifies teacher educators as the nexus where critical AI literacy and HCAI converge and reframes AI literacy as a design practice through which teacher educators structure how preservice teachers first encounter AI. The study offers an evidence-based professional learning model that positions teacher educators as designers and ethical stewards of AI integration and a scalable pathway for preparing future teachers for responsible AI-enabled practice.

1. Introduction

The rapid advancement of artificial intelligence (AI) is reshaping the knowledge, pedagogical practices, and ethical responsibilities expected of educators, creating an urgent need to rethink how future teachers are prepared for AI-enabled learning environments [1,2]. Yet the current education workforce remains largely unprepared to meet this demand. Teachers frequently report feeling ill-equipped to critically evaluate AI tools, integrate them meaningfully into instruction, or guide students ethically in AI-mediated learning contexts [3]. These challenges point to a broader, systemic lack of structured and coherent AI preparation within teacher education programs [3]. As a result, many preservice teachers graduate without opportunities to observe, practice, or receive mentorship in designing AI-integrated instruction or supporting responsible AI use in classrooms [4].

Teacher competence in AI extends far beyond technical tool use. It encompasses ethical reasoning, sociotechnical awareness, transparency, and the intentional design of meaningful learning experiences that align with educational values [5]. Developing teachers' AI literacy, the capacity to understand, question, and shape AI use in ways that support human values and pedagogical goals, is therefore essential for fostering equitable, reflective, and pedagogically sound classroom practices [6,7]. These demands reveal a growing professional preparation gap that cannot be adequately addressed through ad hoc workshops or tool-focused training alone [8]. Instead, they call for systemic, research-informed approaches embedded within teacher education and professional learning infrastructures [9].

Teacher educators play a pivotal role in shaping how future teachers develop the competencies needed for ethical and responsible AI integration. Their responsibilities extend beyond teaching instructional strategies or assessment models; they are increasingly expected to model pedagogical foresight, innovation leadership, and critical engagement with emerging technologies [10]. However, despite widespread recognition of the importance of preparing preservice teachers for AI-enabled teaching, most teacher educators themselves have not received systematic preparation related to AI's ethical, curricular, or instructional dimensions [11]. This misalignment creates a critical paradox: those charged with preparing the next generation of educators often lack the

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institutional support, professional learning opportunities, and design frameworks necessary to engage confidently and critically with AI.

As AI becomes embedded in core instructional practices, including assessment, feedback, personalization, and instructional design, teacher educators' readiness shifts from optional to essential. Their ability to frame AI as a resource for inquiry, creativity, and ethical reflection directly shapes how preservice teachers conceptualize AI's role in teaching and learning [12]. When teacher educators are equipped to model thoughtful, human-centered AI integration, preservice teachers are more likely to enter the profession prepared to use AI critically and responsibly, in ways that prioritize human agency, equity, and professional judgment [4]. Although recent initiatives have aimed to build K-12 educators' AI literacy [13–15], few provide robust, evidence-based models that explicitly support teacher educators in developing these literacies in ways that are contextually grounded and responsive to K-12 classroom realities [9]. Too often, professional learning efforts frame educators as end users or passive recipients of technological solutions, as opposed to professionals who must critically interrogate, adapt, and reimagine AI within their instructional, ethical, and institutional contexts.

A human-centered approach offers a productive alternative to this deficit framing by positioning educators as co-designers of AI-enabled learning environments. Instead of treating AI as a deterministic force, human-centered AI frameworks highlight educator agency, professional judgment, and contextual decision-making. In doing so, these frameworks support deeper forms of AI literacy that extend beyond tool use to include ethical reasoning, pedagogical alignment, and sociotechnical awareness [16]. From this perspective, preparing teacher educators to engage with AI is not simply a matter of skills acquisition, but a design challenge that requires rethinking how professional learning is conceptualized, structured, and sustained within teacher education.

1.1. Purpose

The purpose of this study is to investigate how a human-centered AI in education approach can inform the design of professional learning resources for teacher educators. Specifically, this study is guided by the following research questions:

- RQ1: How do teacher educators integrate AI into their teaching in ways that address human-centered needs, priorities, and challenges?
- RQ2: How can human-centered practices inform the design of professional learning resources that support ethical and pedagogically grounded AI integration?

2. Review of the literature

2.1. Critical AI literacy for educators

Across international policy and research, there is a growing consensus that educators require robust AI literacies and competencies to engage meaningfully with AI in educational contexts [5]. Early conceptualizations of AI literacy primarily emphasized functional and technical skills, such as operating AI tools, crafting effective prompts, or improving efficiency in instructional tasks [17]. More recent frameworks, however, extend beyond technical proficiency to emphasize critical engagement with the purposes, assumptions, and sociopolitical implications of AI in schools [18].

Within this evolving landscape, critical AI literacy has emerged as a multidimensional construct that captures educators' capacity to understand, critique, and reimagine AI in ways that align with pedagogical goals and human values [19]. Critical AI literacy positions educators as reflective professionals who interrogate AI systems as sociotechnical artifacts that are shaped by data, design choices, institutional priorities, and cultural contexts. In doing so, it integrates technical understanding with ethical reasoning and pedagogical judgment. Based on prior scholarship, critical AI literacy encompasses a set of interrelated competencies:

- **Foundational knowledge of AI:** Understanding how AI systems function, including data sources, algorithms, and outputs, as well as recognizing their affordances and limitations [20].
- **Ethics of AI in education:** Examining ethical concerns such as algorithmic bias, data privacy, surveillance, transparency, and accountability in educational uses of AI [21]. This shift reflects an increasing recognition that meaningful AI integration in education requires understanding how AI works and questioning why, for whom, and with what consequences it is used.
- **Pedagogical integration of AI:** Designing instructional practices that use AI to support meaningful learning while maintaining learner agency, creativity, and critical thinking [22].
- **AI implementation, guidelines, and policy:** Interpreting and enacting institutional, district, and national policies that govern AI use in educational contexts [5].
- **Human-centered AI in education:** Positioning educators as active decision-makers who guide AI use in ways that foreground human values, student flourishing, and professional judgment, grounded in principles of human-centered design [23,24].

Together, these competencies move educators beyond surface-level tool use toward a deeper capacity to anticipate and respond to unintended consequences of AI, such as reinforcing stereotypes, promoting over-reliance on automation, or diminishing student creativity [25]. Critical AI literacy thus constitutes both a technical and ethical practice equipping educators to design learning environments that prioritize transparency, equity, and reflective human judgment.

Despite growing agreement on the importance of critical AI literacy, professional learning initiatives remain fragmented and uneven. Many focus narrowly on individual tools or short-term technical skills instead of fostering the conceptual, ethical, and pedagogical understanding necessary for sustained instructional decision-making [26]. Teachers who participate in AI-related coursework or professional development frequently report increased confidence, curiosity, and awareness of AI's instructional potential [13,27]. However, many continue to feel underprepared to address foundational concerns such as data privacy, algorithmic bias, ethical use, and student overreliance on AI [3,28,29]. Systemic barriers, fragmented initiatives, unclear policies, and limited collaborative reflection prevent educators from translating AI knowledge into sustainable classroom practice. These gaps point to the need for professional learning models that move beyond skills-based training toward critical, human-centered approaches to AI literacy. Such models can support educators and teacher educators in questioning the assumptions embedded in AI systems, adapting technologies responsibly to their contexts, and guiding students in developing agency, discernment, and ethical awareness in AI-mediated learning environments.

2.2. Human-centered AI (HCAI) in education: a research and design approach

HCAI has emerged as a response to growing concerns that AI systems, if designed and deployed without sufficient attention to human values, agency, and context, risk undermining trust, equity, and meaningful human decision-making [30,31]. As an interdisciplinary paradigm drawing from human–computer interaction, ethics, learning sciences, and sociotechnical systems theory, HCAI prioritizes the principle that AI should augment human judgment without replacing it, and that humans must retain meaningful control over AI-supported decisions, particularly in high-stakes domains such as education. In contrast to automation-centered or efficiency-driven models of AI adoption, HCAI emphasizes human agency, accountability, transparency, and value alignment [32]. From this perspective, AI systems are not neutral tools; they embody design assumptions, data histories, and institutional priorities that shape how learning, assessment, and student behavior are interpreted [33]. HCAI therefore calls for deliberate attention to who designs AI systems, who controls them, whose values they reflect, and how they redistribute power within educational ecosystems.

HCAI reframes AI as a sociotechnical system that must be intentionally shaped by educators, learners, and institutions [34,35]. Educational decisions such as grading, placement, feedback, and intervention carry significant and long-term consequences for learners, making the delegation of professional judgment to opaque or unaccountable AI systems especially concerning [24]. A foundational principle of HCAI is maintaining humans in the loop across the lifecycle of AI use from design and selection to enactment, monitoring, and revision [36,31]. In educational contexts, however, the generic notion of a human in the loop obscures the distinct roles, responsibilities, and forms of agency that different actors hold. To make this principle pedagogically actionable, we adapt human-in-the-loop into two complementary, role-specific models: student-in-the-loop and educator-in-the-loop. This distinction recognizes that students and educators encounter AI from different positions of authority, expertise, and decision-making responsibility, and that each role requires its own forms of literacy, judgment, and oversight.

Student-in-the-loop refers to students retaining meaningful agency over their own learning processes when AI is involved [37]. Operationalizing this model requires that students develop the literacies needed to interpret, question, and critically evaluate AI outputs when collaborating with AI tools [38]. For example, students may use AI to receive feedback on their writing while still relying on their own judgment and their teacher's feedback before deciding what to revise. They can also cross-reference AI outputs with other resources before incorporating them into their own work.

Educator-in-the-loop refers to educators retaining authority over instructional decisions and being able to intervene in, override, or reinterpret AI outputs as needed [39,40]. Operationalizing this model requires educators to keep their pedagogical goals, content knowledge, and contextual judgment at the center of instructional decision-making when AI is involved [41]. For example, educators may use AI as a drafting tool to generate initial lesson plan ideas and then revise these drafts to align with their disciplinary commitments, student needs, and learning goals. They can also facilitate classroom discussions about AI outputs, positioning students as interpretive authorities who critically examine AI-generated content. In assessment contexts, educators may use AI to refine the wording of feedback they have drafted, while retaining professional responsibility for evaluating student learning. Across these practices, educators establish clear parameters for acceptable and appropriate AI use in their classrooms, modeling the kind of bounded, deliberate engagement with AI they expect of their students.

For teacher education, the emphasis on professional judgment is especially critical [42]. Teacher educators are responsible for teaching instructional strategies, modeling how professionals reason through uncertainty, ethical tension, and competing priorities. When AI systems are presented as authoritative or objective, they risk displacing these forms of professional reasoning [43,44]. HCAI counters this tendency by explicitly valuing interpretation, critique, and contextual decision-making as core professional practices that must be cultivated in preservice teachers.

HCAI is closely aligned with value-sensitive design, which advocates for embedding ethical and social values such as equity, privacy, inclusivity, and cultural responsiveness directly into the design and evaluation of technologies [45]. In educational settings, this alignment is particularly salient given documented concerns about algorithmic bias, discriminatory data practices, and uneven access to AI-enabled resources [46]. Instead of treating ethics as an add-on or compliance requirement, HCAI conceptualizes ethical reasoning as an ongoing, situated practice. Educators are encouraged to ask whether AI tools function correctly and whether they serve educational purposes that promote student flourishing, protect learners' rights, and respect community values [24]. This orientation shifts responsibility away from individual teachers reacting to AI harms after the fact and toward proactive, collective deliberation about acceptable and unacceptable uses of AI in education.

A key contribution of HCAI to educational research is its explicit recognition of AI as a sociotechnical system embedded within broader institutional, political, and cultural contexts [47]. For teacher educators, this perspective is particularly important, as it encourages examination of how AI interacts with curriculum standards, accountability pressures, assessment practices, and school-level policies. HCAI supports forms of professional learning that help educators surface hidden assumptions in AI systems, question whose knowledge is prioritized, and consider how AI may reinforce or disrupt existing inequities. This sociotechnical awareness aligns closely with critical traditions in teacher education that emphasize reflective practice, inquiry-oriented pedagogy, and attention to power and justice in schooling [48,20].

HCAI reframes teacher educators not as intermediaries tasked with transmitting AI knowledge, but as co-designers and ethical stewards who shape how AI is understood, enacted, and contested in educational practice. To operationalize HCAI for teacher education, we synthesized key principles from the HCAI literature in Table 1 and mapped them to concrete applications and examples relevant to teacher educators'

Table 1
HCAI principles and applications for teacher education.

Principle	HCAI Literature (AI systems must...)	Applications & Examples for Teacher Education
Human-in-the-loop	Enhance, not replace, human judgment, enabling users to retain control over decisions and outcomes [36].	Teacher-in-the-loop models in which educators retain pedagogical authority and intervene when necessary [40].
Transparency and Explainability	Provide comprehensible justifications for outputs, allowing users to understand how decisions are made [49].	Integrate explainable AI tools and activities in teacher education coursework to help educators examine how models generate predictions, recommendations, or content [50].
Value-Sensitive Design	Embed ethical and social values, such as equity, privacy, inclusivity, and cultural responsiveness, throughout system design and use [45].	Engage teacher educators in evaluating AI and EdTech tools using structured values-based checklists to assess alignment with educational and community values [18].
Participatory/ Inclusive Co-Design Models	Actively involve diverse stakeholders, especially those affected by AI systems, in design and decision-making processes [51].	Involve teacher educators and K-12 practitioners in the co-design of AI tools, curriculum materials, and professional learning experiences to ensure contextual relevance and inclusivity [52].
Educator Agency & Professional Judgment	Position educators as professionals who can question, adapt, resist, or repurpose AI systems instead of complying with automated outputs [39,4].	Support teacher educators in designing classroom- and institution-level AI policy statements that reflect pedagogical intent, ethical considerations, and professional discretion [28].
Sociotechnical Awareness	Acknowledge that AI systems are embedded within broader social, political, economic, and institutional contexts [47].	Engage teacher educators in analyzing cases of algorithmic bias, data misuse, or inequitable AI design to surface structural implications for teaching and learning [46].
Iterative and Reflexive Practice	Enable systems and practices to evolve through ongoing feedback, reflection, and contextual adaptation [53].	Promote reflective professional learning cycles in which teacher educators iteratively evaluate, revise, and adapt AI-integrated curricula in partnership with diverse stakeholders [54].

professional learning and instructional design practices. This mapping illustrates how abstract HCAI principles such as human-in-the-loop decision-making, transparency, value-sensitive design, and sociotechnical awareness can be translated into actionable pedagogical strategies, policy design activities, and reflective practices within teacher education contexts.

In this study, we adopt HCAI as both a conceptual framework and a design lens to examine how professional learning resources can support teacher educators in developing critical AI literacy. By embedding human-centered principles into curriculum design and professional learning activities, we aim to ensure that AI integration in teacher education remains aligned with educational values, responsive to contextual realities, and grounded in sustained professional judgment.

3. Method

This study employed a design-based research (DBR) method [55,56] to iteratively design, enact, and refine an online professional learning curriculum for teacher educators. DBR was selected for its capacity to generate both theoretical insights and practical design knowledge through sustained collaboration with practitioners and iterative refinement in authentic educational contexts [57]. The research was conducted as part of a teaching innovation project, funded between 2023 and 2025 at a U.S. research intensive university. This multi-phase design project brought together teacher educators, instructional designers, and researchers to co-design, pilot, and revise an online curriculum focused on human-centered and critical approaches to AI in education. This study was approved by the Institutional Review Board (IRB).

Across the study, we engaged in iterative cycles of analysis, design, enactment, reflection, and revision to pursue two complementary aims: (a) developing a scalable, HCAI professional learning curriculum for teacher educators and (b) advancing conceptual understanding of how HCAI principles and critical AI literacy can be operationalized through concrete learning tasks, design scenarios, and reflective activities aligned with teacher educators' instructional responsibilities, institutional contexts, and ethical concerns. The DBR process unfolded across three overlapping cycles (see Fig. 1), with insights from each cycle informing subsequent design decisions and refinements.

Cycle 1: Analysis and problem framing

The first cycle focused on problem analysis and design requirement development. We conducted semi-structured interviews with five teacher educators representing diverse disciplinary areas within a teacher preparation program, including educational technology, literacy, STEM education, and curriculum and instruction. Interview questions explored participants' existing conceptions of AI, perceived opportunities and concerns related to AI use in teacher education, institutional constraints, and aspirations for preparing preservice teachers for AI-enabled classrooms. Analysis of interview data revealed common tensions around ethical responsibility, uncertainty about pedagogical integration, lack of policy clarity, and limited opportunities to model HCAI practices within coursework. These findings informed an initial set of design requirements and guiding principles, grounded in the Critical AI Literacy competencies and HCAI principles outlined in the conceptual framework. In this cycle, teacher educators were positioned as experts whose professional judgment and contextual knowledge shaped the framing of the design problem.

Cycle 2: Prototyping and Co-discovery implementation

Building on the design requirements established in Cycle 1, we developed initial prototypes of online learning modules focused on foundational AI knowledge, ethics and bias, pedagogical integration, implementation and policy, and human-centered futures. To test and extend these prototypes, we conducted an in-person co-discovery workshop that brought together teacher educators, K-12 teachers, and other stakeholders, including educators from a school district implementing a district-wide AI initiative. The workshop was intentionally designed as a human-centered, participatory design experience, emphasizing collaborative inquiry. Participants worked in interdisciplinary teams to engage in structured design challenges addressing AI integration across contexts, including instructional design, assessment practices, professional learning, and institutional guidelines. These activities highlighted the interconnected nature of AI decision-making across K-12, higher education, and teacher preparation. Fig. 2 presents educators collaboratively working on a co-discovery activity related to AI's challenges and opportunities for education. In addition to the workshop, we conducted six co-discovery sessions in which two teacher educators engaged with activity prototypes in their classrooms.

At this stage, we collected design sketches, discussion notes, reflective worksheets, and field notes. Data primarily used during this cycle were field notes. These data captured educators' critical reflections, ethical concerns, and emergent design strategies, revealing both convergent needs (e.g., clarity, flexibility, human oversight) and contextual differences in AI integration priorities.

Cycle 3: Iteration, refinement, and evaluation

Insights from the co-discovery workshop and co-discovery sessions

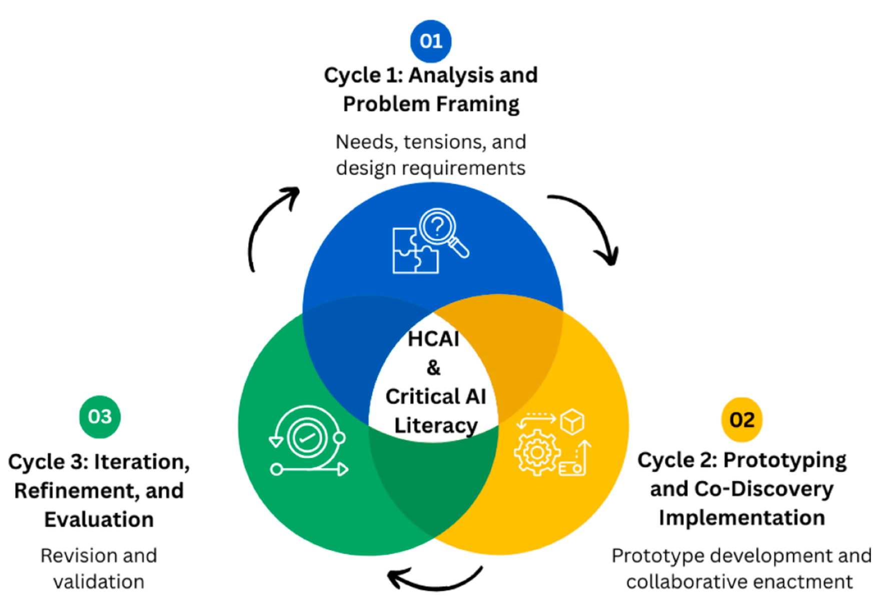

Fig. 1. Design-based research process.

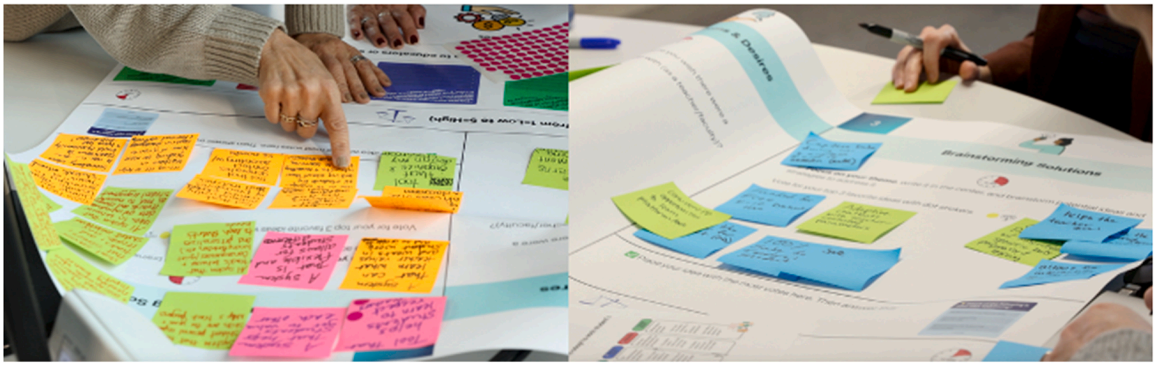

Fig. 2. Teacher educators working on Co-discovery challenges.

revealed opportunities to strengthen the alignment between theoretical framing and educators' lived realities, particularly regarding policy navigation, assessment redesign, and sustaining human agency in AI-mediated instruction. In response, we revised the online modules and accompanying professional learning materials to reflect educator language, classroom constraints, and institutional contexts. To evaluate these revisions, we conducted walkthrough interviews with two teacher educators who engaged with the updated modules. These interviews focused on module clarity, relevance, adaptability across disciplinary contexts, and the extent to which activities supported human-centered decision-making in their contexts. Educators provided feedback on instructional flow, conceptual scaffolding, and practical usability. This cycle culminated in the finalized online curriculum, consisting of five online modules.

3.1. Context and participants

This study was conducted at a large public research university in the Midwestern United States as part of a teacher education program that prepares preservice teachers for both elementary and secondary education. The teacher education program had a learning technology minor that prepared students with the necessary technological skills for emerging technologies they can use in their future classrooms; however, at the time of the study, the program had no formal, program-wide approach to AI integration, creating a timely context for examining how teacher educators conceptualize and enact HCAI practices within their courses.

Participants were recruited through purposive sampling from the teacher preparation program. Eligibility criteria included (a) holding a teacher educator role with responsibility for preparing preservice teachers, (b) some experience with or interest in AI integration, and (c) willingness to participate across the design cycles. Seven teacher educators participated in the data collection process over three DBR cycles. Their professional backgrounds spanned science education, music education, educational technology, literacy education, and social studies, providing a cross-disciplinary perspective on AI integration in teacher preparation. Participants included tenured/tenure-track faculty, teaching professors, and a teaching assistant who taught undergraduate preservice teacher education courses. Table 2 summarizes participants' fields, years of experience, prior AI experience, and the design cycles in which they took part.

Aligning with IRB guidelines, participants completed consent forms and confirmed their participation in the study. To protect participant confidentiality, all names in Table 2 and throughout the findings are pseudonyms.

Teacher educators had varying levels of prior experience with AI, ranging from exploratory experimentation with generative AI tools to more sustained engagement with educational technologies. This variation was intentional and aligned with the study's human-centered orientation, enabling us to examine how educators with different expertise levels articulate needs, tensions, and design priorities related to AI. Throughout the study, teacher educators were positioned as professional partners, contributing their expertise to the iterative development of the online curriculum.

Table 2
Study participants.

Teacher Educator	Role	Field	Years of Experience in Teacher Education	AI Tools Used	Design Cycle
Max	Faculty	Science Education	9	ChatGPT	First Cycle
Claire	Faculty	Music Education	6	ChatGPT	First Cycle
Ella	Faculty	Science Education	9	ChatGPT Copilot Grammarly	First Cycle
Sandy	Faculty	Educational Technology	7	ChatGPT Magic School AI	First Cycle
Vera	Teaching Assistant	Educational Technology	2	ChatGPT Copilot Education Specific AI tools	First Cycle
Kate	Teaching Professor	Literacy Education	5	ChatGPT	Second and Third Cycle
Iris	Teaching Professor	Social Studies	9	Copilot ChatGPT	Second and Third Cycle

3.2. Data sources and collection

Consistent with the design-based research method, we used multiple, interrelated data sources to inform iterative cycles of analysis, design, enactment, and refinement. Data collection was embedded across all three DBR cycles and focused on capturing educators' perspectives, design decisions, and interactions with emerging curriculum materials.

3.2.1. Cycle 1: needs analysis

We conducted semi-structured interviews with five teacher educators. These interviews focused on educators' existing AI knowledge, instructional practices, perceived challenges, and professional learning needs related to AI in teacher education. Interviews lasted approximately 30 minutes and were guided by prompts such as: *"What do you and your students already know about AI?" "How, if at all, are you currently addressing AI in your courses?" "What are the most pressing needs for teacher educators related to AI?"* All interviews were audio-recorded and transcribed verbatim. These data served as the primary basis for problem framing and initial design requirements, informing the selection of content areas, learning goals, and human-centered design principles embedded in the curriculum.

3.2.2. Cycle 2: prototyping and Co-discovery

We collected artifacts and field notes generated during co-discovery sessions in which participants collectively engaged with prototype activities. These artifacts included participant-generated design sketches, discussion notes, and reflective worksheets. In addition, the research team produced field notes documenting participant interactions, emergent questions, and design decisions during each session. Two teacher educators participated in Cycle 2 across 6 co-discovery sessions.

3.2.3. Cycle 3: iteration, refinement, and evaluation

Following revisions to the prototype modules, we conducted walkthrough interviews with two teacher educators who engaged extensively with the updated curriculum. During these approximately one-hour sessions, participants navigated the modules while articulating their interpretations, questions, and critiques. Walkthrough interviews focused on clarity, relevance, adaptability across disciplines, and alignment with authentic teaching contexts, and were guided by prompts such as: "What is your initial reaction to this activity?", "How would this activity fit within your teaching context?", "What would you adapt or change?". All walkthrough interviews were audio-recorded and transcribed verbatim. These interviews were supplemented by annotated module drafts and field notes. We also developed alignment matrices that mapped each curriculum activity to specific HCAI principles and Critical AI Literacy competencies; these matrices functioned both as design artifacts and as analytic tools.

3.3. Data analysis

All qualitative data, including interview transcripts, artifacts, field notes, and walkthrough interview transcripts were analyzed using an iterative process aligned with design-based research principles. Analysis occurred concurrently with design and revision activities, allowing analytic insights to directly inform curriculum development across DBR cycles.

3.3.1. Stage 1: open coding

All research team members independently conducted inductive open coding of interview transcripts and artifacts to identify recurring ideas, needs, and tensions related to AI integration in teacher education. This process generated initial codes by surfacing participants' concerns and priorities without imposing predefined categories [58]. Initial codes reflected issues such as efficiency versus pedagogical authenticity, policy ambiguity, ethical uncertainty, student overreliance on AI, and teacher educators' confidence and agency. Researchers met regularly to compare codes, discuss interpretive differences, and refine code definitions until consensus was reached. Consistent with a constructivist qualitative orientation, we treated analytic consensus as the marker of coding stability.

3.3.2. Stage 2: axial coding and identification of design needs

Codes were organized through axial coding into higher-level categories that reflected both participant-identified priorities and the study's conceptual framework. Categories were iteratively refined to align with HCAI principles (e.g., human-in-the-loop, transparency, value-sensitive design) and Critical AI Literacy competencies (e.g., ethics, pedagogical integration, policy and guidelines). This stage enabled us to trace relationships between teacher educators' expressed needs and the emerging design goals of the curriculum, supporting coherence between empirical findings and theoretical commitments. The convergence of recurring codes across data sources and DBR cycles yielded three interrelated design needs: (a) improving instructional efficiency and effectiveness, (b) modeling responsible and ethical AI integration, and (c) reforming teacher education pedagogies with AI.

3.3.3. Stage 3: alignment matrix and walkthrough validation

In the final stage, we developed an alignment matrix to systematically link the empirical findings to the curriculum design. The matrix served two related functions: it mapped the three design needs to HCAI principles and Critical AI Literacy competencies, and it mapped each curriculum activity to the specific principles and competencies it operationalized. Walkthrough interview data were used to validate, elaborate on, and refine the analytic themes and the corresponding activities. Participant feedback collected during walkthroughs was incorporated into the matrix, supporting ongoing refinement of module structure, language, and adaptability across disciplinary contexts. Throughout the analysis, analytic memos documented emerging insights, design decisions, and theoretical implications. This iterative process generated two complementary outcomes: a practical contribution (the finalized five-module curriculum and accompanying professional development design) and the empirically grounded foundation for the theoretical contribution that illuminates how human-centered approaches can support teacher educators' development of critical AI literacy.Table 3.

3.4. Trustworthiness

We employed several strategies to enhance the trustworthiness and rigor of this study. We followed Lincoln and Guba's [59] framework for credibility, dependability, transferability, and confirmability. Credibility was supported through data triangulation across multiple sources, including semi-structured interviews, participant artifacts, field notes, and walkthrough interviews. The iterative structure of DBR enabled prolonged engagement with participants across 13 months, more than one academic year, allowing us to track how educators' perspectives and design priorities evolved across the three cycles. Analytic rigor was

Table 3
Data sources and analysis methods by design-based research cycle.

Cycle	Phase	Data Sources	Participants	Analysis Methods
Cycle 1	Needs Analysis	Semi-structured interviews (around 30 min, audio-recorded, transcribed verbatim)	5 teacher educators (Max, Claire, Ella, Sandy, and Vera)	Inductive open coding by two researchers; consensus-based refinement; initial code set documenting needs, tensions, and priorities
Cycle 2	Prototyping and Co-Discovery	Participant-generated design sketches, discussion notes, reflective worksheets, field notes	2 teacher educators across 6 co-discovery sessions (Kate and Iris)	Axial coding consolidating Cycle 1 and Cycle 2 codes into three convergent design needs.
Cycle 3	Iteration, Refinement, and Evaluation	Walkthrough interviews (around 60 min) audio-recorded, transcribed verbatim), annotated module drafts, field notes	2 teacher educators (Kate and Iris)	Coding of walkthrough transcripts, construction of alignment matrix linking design needs and curriculum activities to HCAI principles and Critical AI Literacy competencies, validation and refinement of themes

further strengthened through collaborative coding by researchers, regular comparison of interpretations of the data, and the use of analytic memos to document themes and design decisions. A form of design-oriented member checking occurred through the walkthrough interviews. Teacher educators reviewed and responded to prototype modules, and their feedback directly informed analytic refinement and design iteration. While this approach does not constitute member checking in its classical sense (i.e., participants reviewing interpretive accounts of their interviews), it served an analogous function by grounding analytic decisions in participants' lived professional judgments about the design outputs.

Dependability and confirmability were supported through systematic documentation of design cycles, coding procedures, codebook revisions, and the development of alignment matrices linking curriculum activities to HCAI principles and Critical AI Literacy competencies. Memos, design decisions, and inter-coder discussion records were maintained throughout the study to make the analytic path traceable. The research team also engaged in ongoing reflexivity, attending to how our backgrounds in teacher education, learning technologies, and human-centered AI shaped both the design choices and the interpretation of participant data.

Transferability was supported through thick description of the institutional context, participant backgrounds, design cycles, and analytic procedures. Consistent with the analytic orientation of DBR (Cobb et al., 2003), this study emphasizes theoretical and design transferability, offering empirically grounded principles and design insights that can inform professional learning for teacher educators across diverse educational contexts.

4. Results

This results section reports findings from the iterative DBR process and highlights how teacher educators' practices, needs, and challenges informed the development of the online learning curriculum. We organize findings around design-relevant themes that emerged across DBR cycles and directly shaped curriculum decisions. These themes reflected how teacher educators engaged with AI, the tensions they navigated, and the pedagogical priorities they articulated when considering responsible AI integration in teacher education.

4.1. Teacher educators' human-centered needs, priorities, and challenges

Analysis of interview data revealed that teacher educators' considerations around AI clustered around three interrelated goals: (a) improving instructional efficiency and effectiveness, (b) modeling responsible and ethical AI use for preservice teachers, and (c) rethinking teacher education pedagogies in light of AI-enabled teaching and learning.

4.1.1. Improving instructional efficiency and effectiveness

Teacher educators across disciplines described using AI primarily as a thinking partner and tutor. They frequently described AI as a tool for reducing time spent on routine and preparatory tasks and for generating examples. Sandy, reflecting on her educational technology course, shared: *"...what I do with AI is look for examples of if I'm giving this assignment, which misconceptions might students have, or if I need examples that I need to show about a concept, and I only have one or two, I would ask AI to come up with scenarios in this setup, like ways to enrich my work."* Teacher educators also saw the potential to use generative AI tools for lesson planning, generate initial drafts of instructional materials, or outline possible activity structures, which they then refined to better align with course goals, disciplinary norms, and student needs. Participants highlighted AI's potential to support authentic learning experiences by connecting theoretical concepts to real-world classroom contexts. Teacher educators described using AI to generate classroom scenarios, warm-up activities, sample student responses, or discipline-specific prompts that made abstract ideas more concrete. For example, about her music education course, Claire shared: "*I like to start class with a joke, but eventually you run out of like Renaissance music jokes, right? So, that's where AI is helpful.*"

Accessibility was another important dimension of instructional effectiveness. Teacher educators described using AI to adapt instructional materials for learners with diverse language backgrounds, reading levels, or learning needs. For example, Ella, who taught a science education course, reported: *"I want them to be working on labs and worksheets and activities to make them accessible to different, like I said, language acquisition, reading levels. How do we make their activities accessible to everybody?"* AI's value for assisting with assessment and feedback, such as drafting quiz items, rubric language, or feedback comments was also frequently highlighted. Max, reflecting on his science methods class, noted: *"...there's a lot more we can do in our assessment space to hopefully have shorter, more accurate, more informative assessments, kind of real-time formative assessment."*

What participants described as efficiency was, in practice, more nuanced than timesaving. The shared logic running through these accounts was that AI was valued for expanding the educator's range of available choices. The decision about which option to adopt, adapt, or discard remained firmly with the educator. They consistently emphasized that AI should enhance accessibility, feedback, and instructional flexibility while preserving human judgment, contextual adaptation, and professional control over teaching decisions. This pattern pointed to the following HCAI principles, human-in-the-loop decision-making and educator agency.

4.1.2. Modelling responsible and ethical AI integration

Teacher educators viewed modeling responsible and ethical AI use as both a professional responsibility and a pedagogical obligation. Their practices and needs centered on demonstrating transparent AI use, scaffolding ethical decision-making with AI, creating classroom norms and regulations, and fostering reflective engagement. To demonstrate transparent AI use, teacher educators openly shared how they used AI tools in their own teaching and encouraged preservice teachers to disclose and reflect on their AI-assisted work. Vera, regarding her course Educational Technologies for Secondary Education, noted: *"We went through the copyright, we showed them how to properly cite different resources and AI is one example in all the references and citation."*

Teacher educators also valued scaffolding ethical decision-making with AI by guiding students to analyze when and how to use AI responsibly, evaluate potential risks and biases, and make informed choices aligned with professional standards. Vera shared an example: *"We talk about risks to students, to schools, families, and society. So, we had a lot of discussion, and I gave them five different risks to think about, like data privacy, algorithms, bias, those kinds of things. So, we discuss in depth."* Teacher educators reinforced these efforts through creating classroom norms and regulations that promoted accountability and open dialogue about ethical AI use. Finally, educators emphasized the importance of reflective engagement through a continuous process, encouraging preservice teachers to question issues of bias and equity and consider broader societal implications.

What united these practices was that participants treated ethics not as a topic to be covered but as a practice to be enacted in view of preservice teachers. From this perspective, ethical dispositions toward AI were learned less through content delivery and more through watching how the educator handled AI in real time. A second pattern across these accounts was that ethics was distributed across many small, repeated decisions and not concentrated in a single curricular moment, which positioned everyday instructional choices as the primary site of ethical formation. Teacher educators framed ethical AI integration as an ongoing pedagogical responsibility grounded in critical reflection, contextual judgment, and attention to broader societal implications. These priorities closely aligned with HCAI principles related to transparency and explainability, value-sensitive design, and sociotechnical awareness.

4.1.3. Reforming teacher education pedagogies with AI

Integrating AI into teacher preparation prompted a reevaluation of pedagogical and assessment practices to preserve agency, authenticity, and meaningful learning. Teacher educators emphasized the need for AI-resistant assessments that prioritize justification and reflection over AI-generated output. They recognized that traditional essays and open-ended writing tasks were increasingly susceptible to reliance on AI tools. Therefore, they focused on authentic, process-based assessments grounded in personal experience and critical thinking. Participants also emphasized iterative practices, in which summative assessments were replaced with smaller, scaffolded tasks that invited students to engage critically with AI output at multiple stages. For example, Sandy noted: *"…AI is going to look different in a couple of years, but I want them to be aware of what can be done and have thoughtful guidance for their students and their classrooms."*

Taken together, these shifts pointed to a broader reframing of teacher education itself, reflecting HCAI's emphasis on educator agency, participatory co-design, and iterative reflective practice. They underscored the importance of designing learning environments in which AI supported human capabilities such as critical thinking, creativity, adaptability, empathy, and problem-solving.

4.1.4. Designing teacher professional learning with HCAI

Building on insights from teacher educators' practices and articulated needs, the DBR process shifted from documenting AI use toward translating human-centered principles into concrete professional learning design features. Across DBR cycles, findings revealed that teacher educators did not seek prescriptive guidance on how to use AI. They wanted to explore learning experiences that can support educators' professional judgment, ethical implementation, and pedagogical decision-making depending on their teaching context. Collectively, these findings informed the design of the online curriculum by translating teacher educators' articulated needs into concrete learning activities grounded in HCAI principles, including human-in-the-loop decision-making, transparency, educator agency, sociotechnical awareness, and reflexive practice. The remainder of this section presents each activity together with the design need(s) it responds to and the HCAI principle(s) it operationalizes, making the chain from participants' articulated needs to instructional design choices explicit.

4.2. Designing for educator agency and human-in-the-loop decision-making

A central HCAI design implication was the need to center educator agency by ensuring that AI-supported activities consistently reinforced human-in-the-loop decision-making. Teacher educators perceived a risk that AI tools, if not intentionally framed, could unintentionally displace pedagogical reasoning or be interpreted by preservice teachers as authoritative sources. To address this concern, the curriculum activities were deliberately designed so that AI outputs were treated as provisional and subject to human interpretation and judgment. For example, to equip educators with the foundational knowledge of AI, we designed a hands-on activity using Google's Teachable Machine, a web-based tool that enables users to train simple machine-learning models through data collection, labeling, training, and testing without coding. By creating their own image-based models, educators explored how AI systems learn from human-provided data, make predictions, and improve through iteration, while also observing how errors and misclassifications emerge from data choices and model limitations. Fig. 3 provides an example of the Teachable Machine activity interface, illustrating how educators interact with data collection, model training, and output interpretation during the activity.

During the walkthrough interviews, teacher educators emphasized the importance of educator agency. When the activity was shown to her, Kate explained: *"This is exactly what I want... I want AI to make my students more critical in their thinking and less stressed about the mechanics. They need support, like a little coach, for the endless questions about teaching. No one person has all the right answers, but AI can help them*

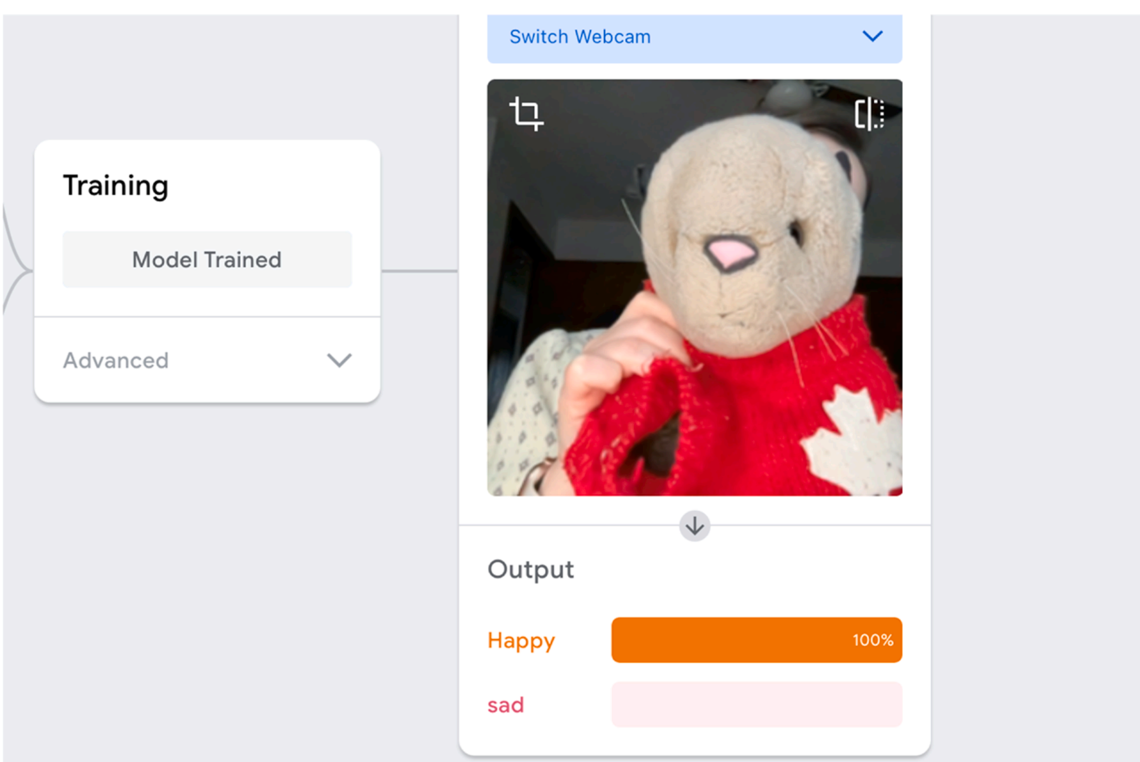

Fig. 3. Teachable machine interface used for model training.

explore possibilities." This perspective illustrated the critical value educators placed on maintaining human judgment as the central driver of instructional decision-making, treating AI as a tool for exploration and support.

This activity directly emerged from teacher educators' emphasis on preserving human judgment while reducing cognitive and procedural burdens associated with AI-supported teaching. The design intentionally facilitated educator interpretation, experimentation, and critique instead of treating AI systems as authoritative decision-makers. In this way, the activity operationalized participants' desire for AI to function as a supportive thinking partner that expanded pedagogical possibilities without replacing professional reasoning. This activity addressed the design needs of improving instructional efficiency and effectiveness, and reforming teacher education pedagogies with AI. Educators could directly inspect how AI works and makes decisions about its use, enacting HCAI principles of human-in-the-loop oversight and transparency.

4.2.1. *Making AI decisions visible: transparency and explainability in practice*

To operationalize the HCAI principles of transparency and explainability in the Ethics topic, we designed the Ethical Principles Cards activity. Twelve ethical principles cards were designed to scaffold reflection on AI integration decisions. Each digital flip card presents a reflective question on the front aligned with a specific ethical principle and a concrete classroom example with a corresponding action prompt on the back. For example, the Bias Awareness card asks, *"Could this AI integration potentially treat some individuals unfairly?"* and illustrates this concern through an example in which an AI tutoring system generates shorter responses to Spanish-accented speech. The accompanying action step prompts educators to test AI tools using inputs from different dialects, genders, or cultural contexts and to critically examine who is represented fairly and who may be misjudged. This design moves beyond identifying bias toward actionable strategies for transparent evaluation and responsible classroom use, as illustrated in Fig. 4.

Teacher educators perceived transparency and explainability as essential for helping preservice teachers engage critically with AI without developing fear or overreliance. For example, during walk-through interviews, Iris, regarding her social studies methods course, explained, *"[The ethical principles cards] is something I hugely want. I also want my students to learn how to use AI for efficiency, but also to maintain critical [thinking]. Because I don't want them to be scared of AI."* The design of the Ethical Principles Cards was closely informed by teacher educators' repeated calls for practical and discussion-oriented approaches to AI ethics. To move beyond presenting ethics as abstract policy knowledge, participants emphasized the need for approaches that would help preservice teachers recognize ethical tensions within authentic classroom decisions. Accordingly, the activity was designed to translate complex concepts such as bias, representation, and accountability into actionable and contextually grounded reflection prompts. The Ethical Principles Cards responded directly to the design need of modeling responsible and ethical AI integration and applied the HCAI principles of transparency and explainability, value-sensitive design, and sociotechnical awareness by giving teacher educators classroom-grounded artifacts through which ethical reasoning could be made visible to preservice teachers.

4.2.2. *Case-based exploration of values in AI-integrated teaching*

Guided by value-sensitive design principles that foreground ethical, social, and pedagogical values in technology use, the curriculum operationalized these commitments through a set of real-world classroom- and school-based cases. Through these cases, educators examined how teachers navigated issues of responsibility and instructional purpose, shifting the focus from whether AI was used to how and why it was integrated into teaching practice. By analyzing authentic instructional decisions and reflecting on the values that shaped them, educators were encouraged to identify ethical dilemmas, pedagogical trade-offs, and moments where human judgment was essential. Fig. 5 presents an example of a case interface designed to support value-driven reflection and discussion.

Teacher educators perceived this case-based approach as embedding value-sensitive reasoning directly into instructional analysis. For

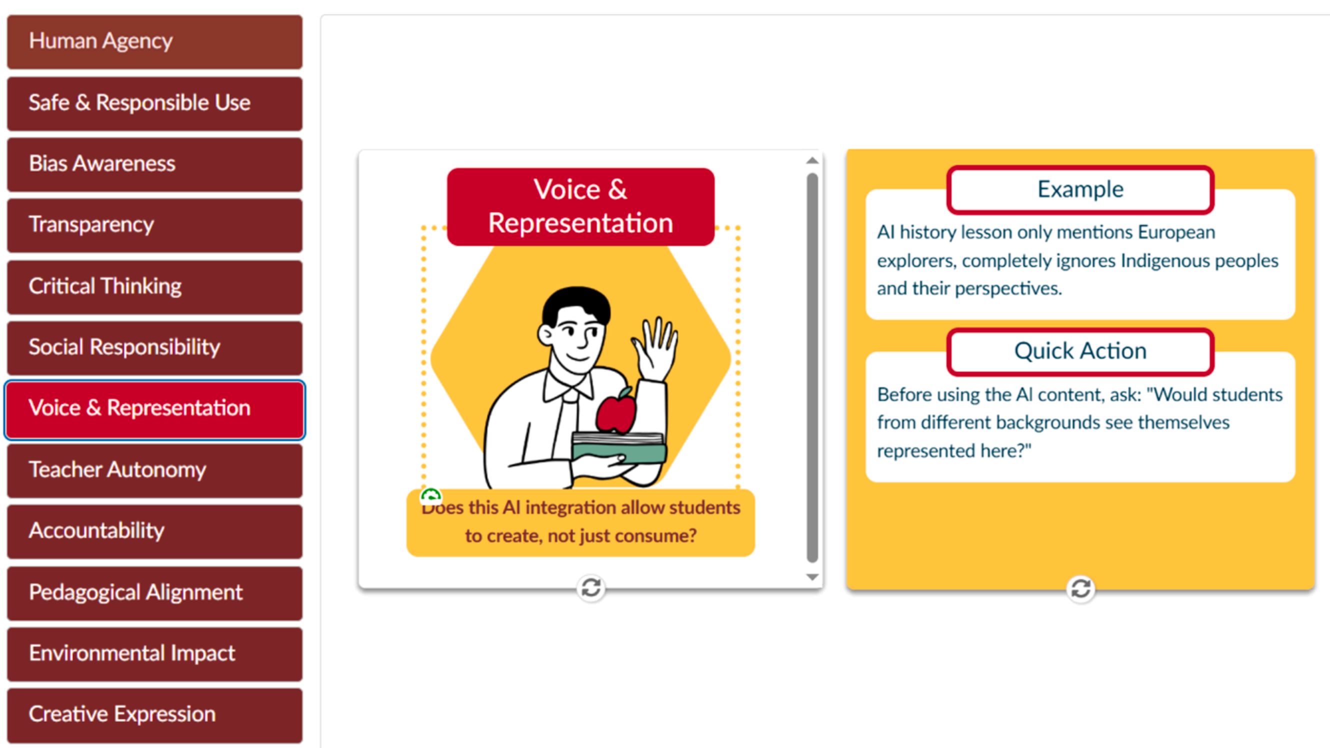

Fig. 4. Example of an ethical principles card illustrating bias awareness.

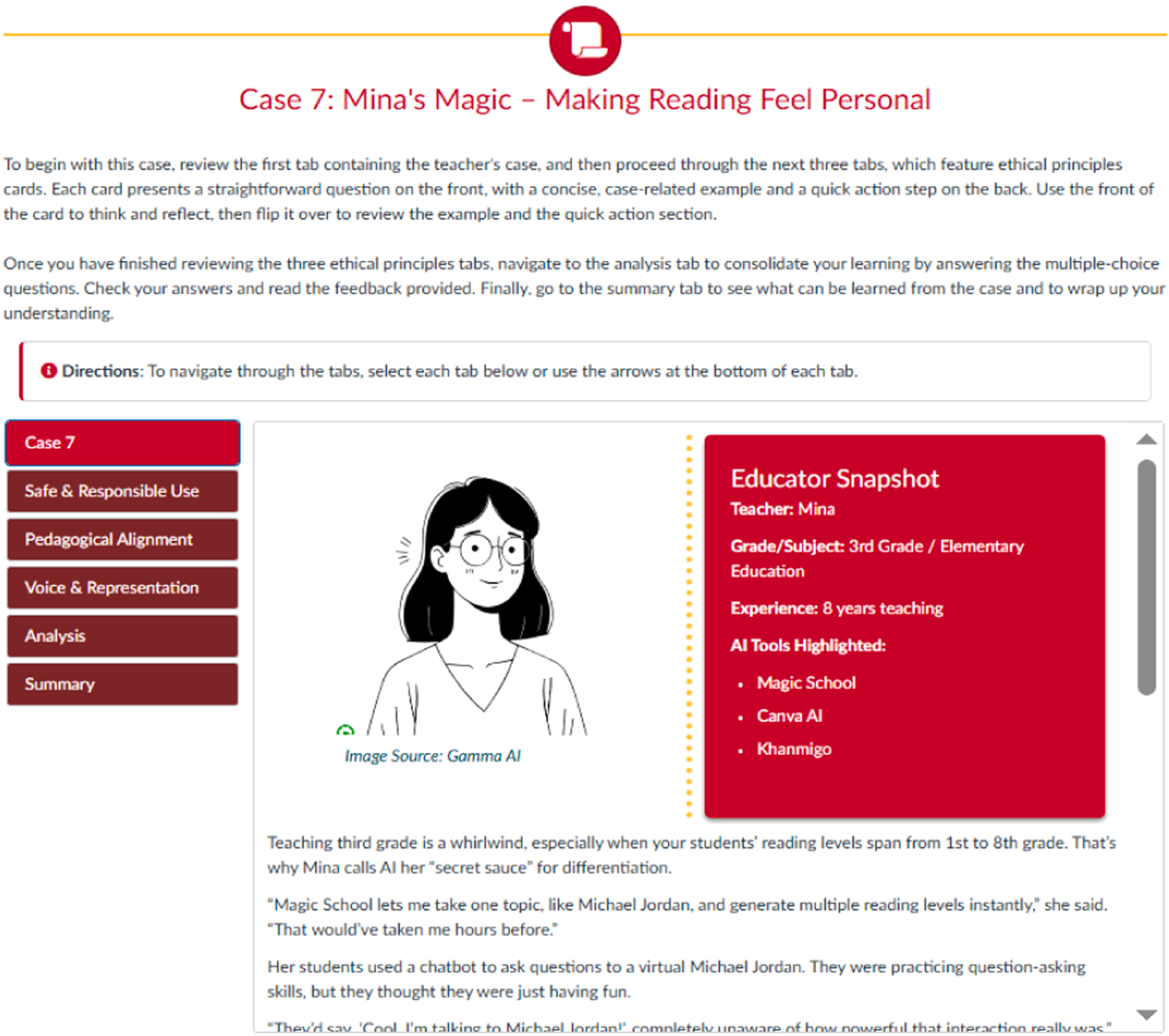

Fig. 5. Example of the interactive case-based learning interface.

example, during walkthrough interviews, Kate explained, *"These cases are really exciting, and I think they can hugely help... I do have students who maybe aren't going to teach traditional academic classes like music, and physical education, and family and consumer sciences... because those teachers sometimes feel like they don't get a lot of cases that relate to their subject areas... But they really need."* This case-based structure echoed teacher educators' concern that AI integration should remain pedagogically intentional, aligning technology use with human values, learning goals, and professional judgment. Participants consistently emphasized that instructional goals, student needs, and disciplinary contexts must precede decisions about AI use. Therefore, this activity underscored the importance of exposing preservice teachers to diverse, value-informed approaches to AI use.

4.2.3. Co-designing classroom pedagogy with AI

The curriculum operationalized co-design through a hands-on lesson planning activity in which educators collaborated with generative AI while retaining pedagogical authority. In the Co-Design a Lesson with AI activity, participants first articulated their instructional intentions, learning goals, pedagogical approaches, and assessment strategies before engaging with any AI tool. This sequencing ensured that pedagogical values and professional judgment guided subsequent design decisions. Educators then compared their self-designed lesson plans with AI-generated alternatives, critically examining points of alignment, tension, and omission. Through iterative refinement, participants adapted, revised, or rejected AI suggestions based on ethical considerations, equity concerns, and the specific needs of their students and instructional contexts. This process positioned AI as a design partner instead of a directive authority and reinforced educators' lived expertise as central to instructional decision-making. Fig. 6 illustrates an example interface from the co-design activity.

Teacher educators viewed this approach as essential for helping preservice teachers understand how AI could be integrated in ways that aligned with pedagogical goals. For example, during walkthrough interviews, Iris explained, *"This perfectly aligns with my class when they create lesson plans... I want to introduce the idea, okay, you can use AI for this particular assignment...show me, okay, this is the research that we use using artificial intelligence and this is how we analyze that."*

The activity was developed in response to concerns that preservice teachers may over-trust AI-generated responses without critically

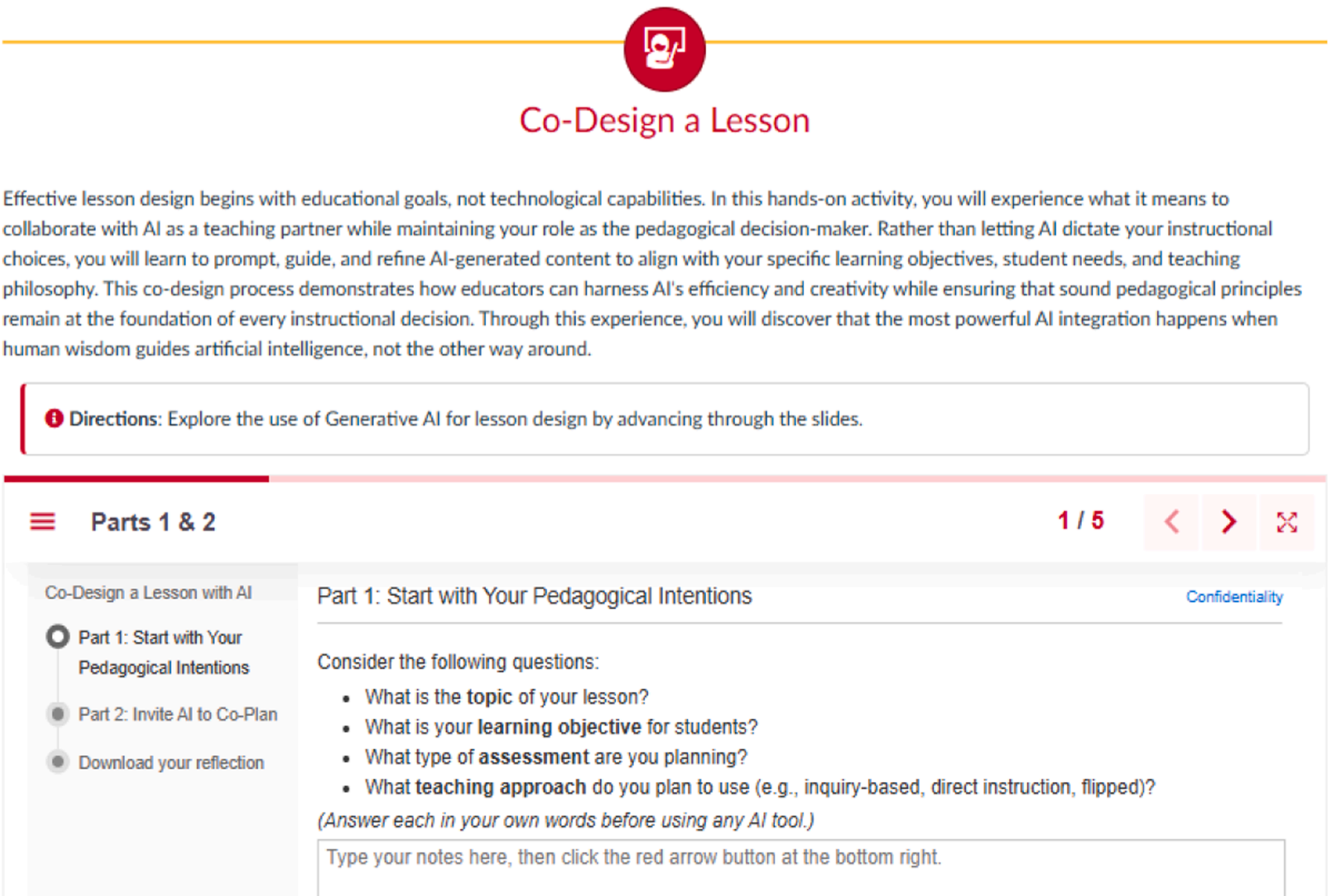

Fig. 6. Example interface from the Co-design lesson planning activity.

examining their assumptions or limitations. The design of the activity emphasized the importance of critique, disagreement, and counterargument as essential learning practices. This approach fostered the development of professional judgment by treating ethical questioning as an active pedagogical responsibility. This activity applied the HCAI principles of educator agency, participatory co-design, and human-in-the-loop decision-making by requiring educators to articulate their pedagogical goals before engaging any AI tool, so that professional judgment structured educators' AI use.

4.2.4. *Educator agency and professional judgment*

The curriculum emphasized educator agency in AI-integrated teaching contexts, situating educators as professionals who could make intentional decisions about AI use. Findings indicated that strengthening educators' capacity to apply informed judgment when engaging with AI was essential, particularly when AI-generated outputs risked being interpreted as authoritative. This need was addressed in an activity that engaged teacher educators in simulated professional dialogue with Copilot, analyzing its responses to ethical questions related to bias, privacy, accountability, and pedagogical impact, and explicitly challenging its reasoning when they disagreed. Through the activity, educators were prompted to interrogate assumptions, identify oversimplifications, and articulate counterarguments grounded in educational values and classroom realities. Fig. 7 illustrates the Interview Copilot Activity.

Regarding critiquing AI outputs and maintaining the teacher's role, Kate stated: "AI has created this. You still have to have the human element... As a teacher, you still have to teach the lesson. AI is not teaching it for you." This activity responded to the design needs of modeling responsible and ethical AI integration and reforming teacher education pedagogies with AI and operationalized the HCAI principles of educator agency and professional judgment and transparency and explainability by treating AI outputs as objects of professional critique.

4.2.5. *Designing for sociotechnical awareness and policy engagement*

Findings revealed that teacher educators often felt unprepared to navigate the policy and governance dimensions of AI use, particularly when institutional guidance was unclear or continued to evolve. In response, the curriculum was designed to connect classroom-level AI decisions to broader sociotechnical systems, including district policies, institutional expectations, and societal implications. The curriculum engaged educators in a step-by-step process for developing classroom AI use policies grounded in their own instructional contexts. Instead of relying on pre-written templates, activities prompted educators to analyze real-world scenarios, identify risks and responsibilities, and articulate guiding principles that reflected their pedagogical values and community expectations. Fig. 8 illustrates the example activity.

Teacher educators perceived these activities as critical for establishing clarity, accountability, and professional judgment around AI use in educational settings. For example, during walkthrough interviews, Kate explained, *"I set guidelines for when and how and why we use AI… and when I don't prefer it. So, they should also learn how to define their classroom policies as well. This would be a great source for them."* Iris also emphasized a similar concept and mentioned, *"If we embrace [AI] instead of trying to hide from it, it is better, and I think we have the accountability to expose our teacher candidates and tell them what is right, what is not, and how it can be so helpful."* These reflections illustrated the critical value educators placed on preparing preservice teachers to navigate AI through clear guidance and ethical responsibility.

This activity responded to the design needs of modeling responsible

Fig. 7. Interview copilot activity interface.

and ethical AI integration and reforming teacher education pedagogies with AI, and enacted the HCAI principles of sociotechnical awareness, educator agency, and human-in-the-loop decision-making by positioning policy as an artifact of professional reasoning.

4.2.6. Iterative and reflexive practice

The importance of providing educators with ongoing opportunities to reflect on their learning aligned with HCAI frameworks that emphasize iterative and reflexive practice as essential to responsible AI integration. Accordingly, the curriculum was designed to embed reflexive practice throughout the learning experience instead of confining reflection to a single endpoint. Following each activity, educators engaged with structured reflection cards consisting of one to three tabs tailored to the activity's learning goals. These cards prompted high-level reflection on what was learned, how insights translated to educators' own instructional contexts, and how AI-related decisions might be refined over time. Reflections were completed through embedded LucidChart boards, where participants shared their responses and design artifacts, engaged with peers' contributions, and iteratively refined their thinking through collective sense-making. Fig. 9 illustrates an example reflection card used to support reflexive practice.

Teacher educators perceived this structured reflection as critical for preserving human judgment and intentionality in AI-supported learning. For example, during walkthrough interviews, Kate explained, *"On lesson plans, I tell them to absolutely use AI, and then I set limits too on their personal reflections. I don't let them use AI on that, just because I want their personal thinking... This approach is a valuable way to have them reflect on the AI-generated content."*

The emphasis on structured reflection directly reflected participants' concerns about preserving intentionality, critical thinking, and human agency within AI-supported learning environments. Teacher educators consistently distinguished between tasks that might appropriately involve AI support and moments that required independent reflection and pedagogical reasoning, treating reflexivity as a pedagogical practice that safeguarded human agency. Consequently, reflection was embedded throughout the curriculum as an ongoing practice of examining, refining, and humanizing AI integration, instead of an add-on at the end of the experience. The reflection cards responded directly to the design need of reforming teacher education pedagogies with AI, and operationalized the HCAI principles of iterative reflective practice and educator agency by making reflection a recurring site of professional judgment within instruction.

4.2.7. Synthesis: Mapping HCAI Principles Across the Curriculum

To synthesize how HCAI principles were operationalized across the curriculum, we mapped each of the seven curriculum activities to their

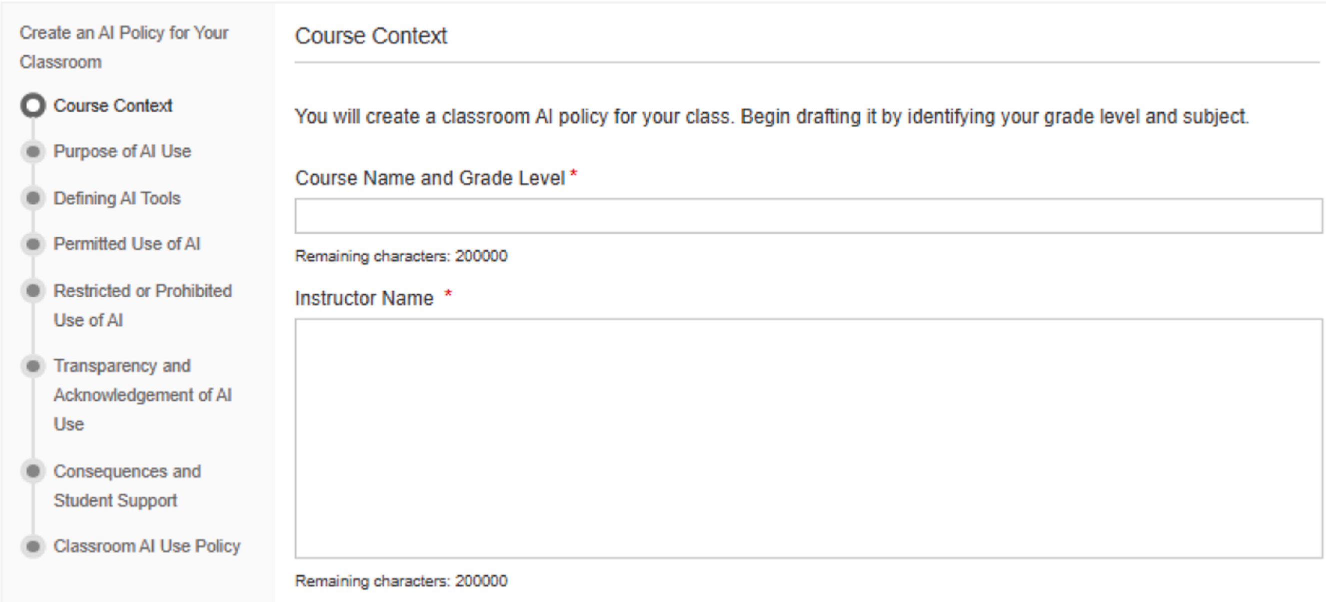

Fig. 8. Classroom AI use policy design interface.

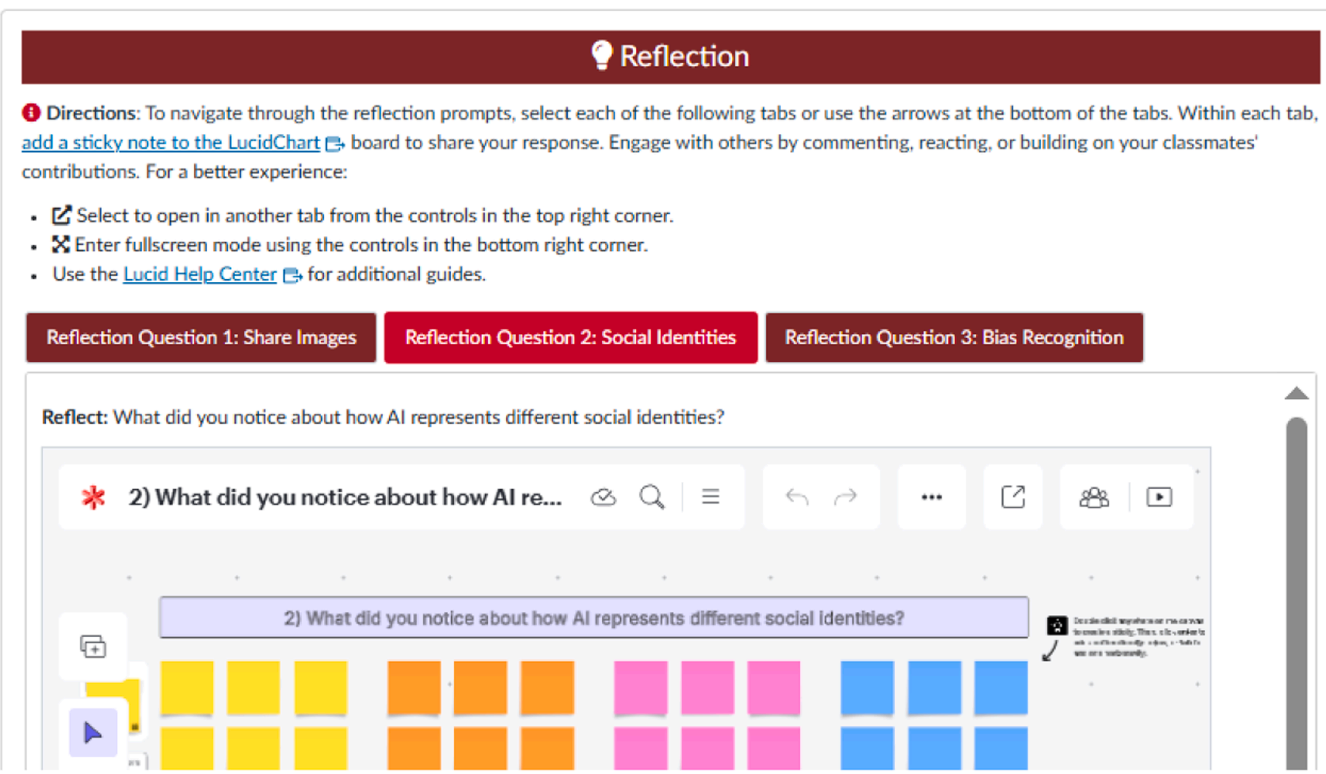

Fig. 9. Example reflection card with an embedded collaborative board.

corresponding module, HCAI principles, and design needs identified in the findings. Each activity is directly traceable to the design needs and priorities teacher educators articulated, such as improving instructional efficiency and effectiveness, modeling responsible and ethical AI integration, and reforming teacher education pedagogies with AI, illustrating the curriculum was a direct response to practitioners' expressed priorities. Table 4 shows how HCAI principles were translated into activities reflecting teacher educators' professional realities.

5. Discussion

Teacher educators play a central role in cultivating future teachers' pedagogical reasoning and critical dispositions toward technology [60]. However, research shows that many feel underprepared to address AI's ethical, pedagogical, and curricular implications [11], reflecting wider concerns about uneven institutional capacity in higher education. This study contributes to the scholarship on AI in teacher education by empirically connecting critical AI literacy and HCAI for designing professional learning for teacher educators. While prior research [61] has

Table 4
Curriculum activities mapped to modules, HCAI principles, and design needs/priorities.

Activity	Module/Activity-Topic Description	HCAI Principles Applied	Design Needs/Priorities
Teachable Machine	Module 1: Foundational Knowledge of AI How AI learns from data, makes predictions, and produces error	Human-in-the-loop, Transparency & Explainability	Modeling responsible and ethical AI integration
Ethical Principles Cards	Module 2: Ethics of AI in Education Bias, privacy, accountability, representation	Transparency & Explainability, Value-Sensitive Design, Sociotechnical Awareness	Modeling responsible and ethical AI integration
Interactive Case-Based Learning	Module 3: AI's Pedagogical Integration Examining AI use through discipline-specific classroom cases	Value-Sensitive Design; Educator Agency & Professional Judgment, Sociotechnical Awareness	Modeling responsible and ethical AI integration, Reforming teacher education pedagogies with AI
Co-Design a Lesson with AI	Module 3: AI's Pedagogical Integration Lesson planning with AI as a critical design partner	Participatory & Inclusive Co-Design, Educator Agency & Professional Judgment, Human-in-the-loop	Improving instructional efficiency and effectiveness, Reforming teacher education pedagogies with AI
Interview CoPilot	Module 2: Ethics of AI in Education Professional dialogue analyzing and contesting AI-generated ethical claims.	Educator Agency & Professional Judgment, Transparency & Explainability	Modeling responsible and ethical AI integration,
Create your AI Policy	Module 4: AI Implementation & Guidelines Scenario-based development of classroom AI governance	Sociotechnical Awareness, Educator Agency & Professional Judgment, Human-in-the-loop	Modeling responsible and ethical AI integration, Reforming teacher education pedagogies with AI
Reflection Cards	Embedded across all modules Structured prompts and peer exchange via collaborative board	Iterative & Reflexive Practice, Educator Agency & Professional Judgment	Reforming teacher education pedagogies with AI

increasingly called for educators to develop AI literacy that extends beyond technical proficiency (e.g., ethics, bias, pedagogical alignment), much of this work remains abstract or focused on practicing teachers. Our findings demonstrate how these frameworks can be operationalized in teacher education contexts, where teacher educators play a formative role in shaping how future teachers interpret, evaluate, and enact AI in schools.

5.1. Critical AI literacy as a design-oriented professional capacity

The findings reinforce and extend conceptualizations of critical AI literacy as a multidimensional professional capacity that integrates technical understanding, ethical reasoning, pedagogical judgment, and sociotechnical awareness. Consistent with prior frameworks (e.g., [18]), teacher educators in this study did not conceptualize AI literacy as mastery of tools or algorithms. Instead, they emphasized the need to reason with AI under uncertainty: deciding when AI use is pedagogically justified, how to surface risks and biases, and how to support students to critically engage with AI outputs.

Importantly, this study extends critical AI literacy scholarship in two ways. First, it positions teacher educators as a distinct population whose AI literacy demands cannot be inferred from frameworks developed for practicing K-12 teachers. Existing scholarship tends to address educators as a single category, assuming that AI literacy developed for K-12 contexts will diffuse upward into teacher preparation; the findings here suggest the opposite, since teacher educators occupy a curricular position that requires distinct forms of professional learning. Second, the study reconceptualizes critical AI literacy as a design practice for teacher educators, not a checklist of competencies. Teacher educators are not merely consumers or evaluators of AI tools; they design the learning environments, assessments, norms, and policies through which preservice teachers first encounter AI. From this perspective, the five critical AI literacy competencies function less as standalone outcomes and more as resources teacher educators draw on when making instructional design decisions under pedagogical, ethical, and policy uncertainty.

The three convergent design needs identified in this study (improving instructional efficiency and effectiveness, modeling responsible and ethical AI integration, and reforming teacher education pedagogies with AI) demonstrate how the five critical AI literacy competencies are enacted within teacher educators' instructional design work. Each design need cuts across several competencies, with no one-to-one correspondence between them. Together, they illustrate how critical AI literacy becomes actionable when embedded in instructional design tasks beyond being treated as abstract knowledge or isolated ethical discussions.

The study results align with recent critical work arguing that AI literacy must be cultivated through authentic, contextually grounded practices, without relying on decontextualized skill training [62,26]. By situating AI literacy within teacher educators' instructional decisions for lesson planning, assessment redesign, modeling transparency, and policy interpretation, this study demonstrates how critical AI literacy can be enacted as a form of professional judgment, not a checklist of competencies.

5.2. HCAI as a pedagogical and epistemic framework

The study extends HCAI scholarship in two ways. First, it shifts HCAI's locus of application from system design to professional learning. HCAI frameworks have traditionally formulated principles such as transparency, explainability, and human-in-the-loop control as design properties of AI systems themselves [31]. The findings here suggest that these principles also need to be cultivated as pedagogical commitments within teacher education, since the humans who are supposed to remain in the loop cannot exercise that role without sustained professional learning that develops their capacity to interpret, challenge, and contextualize AI outputs. Second, the study positions HCAI as a normative orientation for teacher educators. This perspective, principles like transparency or educator agency are not technical features to be engineered but professional dispositions to be developed, modeled, and made visible across the everyday instructional decisions teacher educators make.

Across DBR cycles, teacher educators consistently resisted automation-centered narratives that frame AI as an authoritative or efficiency-driven solution. Instead, they articulated a need for learning experiences that preserve educator agency, emphasize ethical deliberation, and make AI decision-making visible. These priorities closely align with HCAI's emphasis on augmenting human judgment and maintaining meaningful human oversight in high-stakes domains such as education.

By embedding HCAI principles into learning tasks (e.g., evaluating AI outputs, interrogating bias, drafting contextual policies), the curriculum reframes AI not as a neutral tool but as a sociotechnical system shaped by values, data, and institutional constraints. This reinforces arguments in the HCAI literature that ethical AI in education depends on both better algorithms and educators' capacity to interpret, challenge, and contextualize AI use [24,39].

5.3. Teacher educators as the nexus of critical AI literacy and HCAI

A key theoretical contribution of this study is its articulation of teacher educators as the nexus where critical AI literacy and HCAI converge. Existing literature often treats teacher educators as a secondary audience for AI professional development or assumes that AI-related competencies will diffuse upward from K-12 contexts. The findings here challenge that assumption by showing that teacher educators occupy a distinct professional position: they are at once users of AI in their own teaching, models of AI use for preservice teachers, designers of curriculum and assessment, and interpreters of the institutional and ethical frameworks that govern AI in education [10]. This combination of roles requires its own forms of professional learning and makes teacher educators the point where HCAI commitments and critical AI literacy competencies meet in instructional practice.

The iterative DBR design also surfaced insights that no single cycle could have produced on its own. The first cycle, focused on needs analysis, made visible that what teacher educators valued under the language of efficiency was the preservation of their interpretive authority over AI outputs, a finding consistent with HCAI's emphasis on maintaining meaningful human oversight in high-stakes decision-making [31]. The second cycle, through prototype testing and co-discovery, revealed that ethical AI integration was treated by participants as a performed practice in view of preservice teachers, not as content to be delivered, extending arguments that AI preparation cannot be reduced to skills-based training [8,26]. The third cycle, through walkthrough refinement and alignment-matrix work, made visible that AI-resistant pedagogy is concerned less with assessment redesign alone [33] and more with preserving the epistemic conditions under which professional reasoning can be cultivated. Together, these insights suggest that AI literacy for teacher educators is enacted not as a knowledge inventory but through the design choices through which they structure how preservice teachers encounter AI.

6. Conclusions

This design-based research study investigated (a) the human-centered needs, priorities, and challenges teacher educators identify as they engage with AI integration in teacher preparation, and (b) how HCAI and critical AI literacy can be translated into professional learning resources that support ethically grounded and pedagogically meaningful AI integration. Across iterative cycles of inquiry, co-discovery, and refinement, the study surfaced a consistent message: teacher educators are not primarily seeking how-to training on emerging tools but design supports that strengthen professional judgment under uncertainty, surface risks and limitations transparently, and align AI-mediated practices with educational values and institutional expectations. Reading across the cycles together, the findings position teacher educators as the nexus where HCAI commitments and critical AI literacy competencies meet and suggest that AI literacy for teacher educators is best understood as a design practice through which they shape how preservice teachers encounter AI.

7. Design implications for professional learning

Findings of this study point to three design priorities for professional learning aimed at teacher educators, each responding to one of the design needs identified in this study. First, professional learning must be scaffolded and flexible, enabling teacher educators to adapt AI literacy work across disciplinary contexts, program structures, and varied levels of prior AI experience. Second, teacher educators need ongoing guidance on ethics, policy, and responsible use that helps them navigate uncertain institutional terrain and model transparency for preservice teachers. Third, to reform teacher education pedagogies with AI, professional learning should emphasize educator-in-the-loop activity designs that make AI decision points visible and preserve the centrality of professional judgment. These activities should require educators and preservice teachers to interrogate AI outputs, justify use decisions, document limitations, and reflect on consequences, thereby resisting automation-centered narratives that position AI as authoritative or neutral.

8. Recommendations for teacher education programs and researchers

Building on the design priorities above, we offer three recommendations for teacher education programs. First, programs should embed critical AI literacy as an explicit and sustained design goal across coursework and field experiences, not as an add-on workshop topic. This includes supporting teacher educators to model transparent AI use, redesign assessments for authenticity and reasoning, and cultivate preservice teachers' capacity to evaluate and contextualize AI output. Second, programs should support teacher educators in developing shared guidelines and policy-facing language that reflect local values and constraints, defining policy engagement as a form of professional agency instead of top-down compliance. Third, programs should establish structures for ongoing collegial inquiry, such as communities of practice, co-design studios, and iterative curriculum pilots, so that AI literacy development remains responsive as tools, policies, and classroom realities shift.

For research, this study points to the need to move beyond questions of whether teachers adopt AI and toward investigations of how educator agency is supported or constrained within AI-mediated educational systems. Future research should examine how HCAI-informed professional learning scales across institutions, how teacher educators' design decisions shape preservice teachers' classroom practices over time, and how institutional policies interact with educators' ethical commitments and pedagogical priorities. Longitudinal research is also needed to understand whether and how critical AI literacy becomes durable, especially given evidence that, without continued scaffolding and alignment to curricular and policy infrastructures, educators may revert to instrumental AI uses that prioritize efficiency over learning [63].

9. Limitations

This study has several limitations that should be considered when interpreting the findings and the transferability of the proposed professional learning model. First, the work was conducted within a single institutional context and with a relatively small group of teacher educators and partners; while participants represented varied disciplinary perspectives, the design needs identified here may manifest differently in other teacher preparation programs with distinct governance structures, licensure requirements, student demographics, and policy environments. Second, as is typical in design-based research, the study prioritized iterative refinement and ecological validity over broad representativeness. The resulting curriculum and design principles are intended to be theoretically transferable, not statistically generalizable, and further research is needed to test how these design features function across additional sites and conditions.

Third, the study emphasized educators' perceptions of relevance, usability, and alignment with human-centered and critical aims, while also attending to downstream impacts on preservice teachers' learning or classroom enactment. Although the curriculum is explicitly designed to cultivate critical AI literacy competencies and educator-in-the-loop decision-making, we did not assess whether preservice teachers subsequently adopted these practices in field placements or early career teaching, nor did we examine longer-term effects on instructional quality, equity outcomes, or student learning. Finally, the rapid evolution of generative AI tools, district policies, and institutional guidance limits the stability of any fixed set of examples or platform-specific activities. While the curriculum was designed around durable principles (e.g., transparency, agency, value-sensitive judgment), continued updating and local adaptation will be necessary to ensure ongoing relevance as sociotechnical conditions change.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used generative AI tools to assist with language editing and clarity (e.g., grammar, wording, and readability). The authors carefully reviewed and edited all content and take full responsibility for the accuracy, integrity, and originality of the work. No generative AI tools were used to generate research data, analyses, or scholarly interpretations.

CRediT authorship contribution statement

Evrim Baran: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Melis Dilek:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Melika Ziba:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Xiao Xiao:** Writing – review & editing, Visualization, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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